
A Median Perspective on Unlabeled Data for Out-of-Distribution Detection

Anonymous Author(s)

Affiliation

Address

email

Abstract

Out-of-distribution (OOD) detection is fundamental to reliable machine learning under open-world deployment, where test inputs may originate outside the label space represented during training. Unlabeled in-the-wild data provides a practical source of deployment-relevant evidence, but it also creates a central methodological difficulty: the available pool is a contaminated mixture of in-distribution (InD) and OOD samples whose individual memberships are unobserved. We introduce Medix, a median-centric framework that converts this unlabeled mixture into supervision for OOD detection by extracting candidate outliers and training a binary detector with those candidates and labeled InD data. Medix uses the median as a robust estimator of central gradient behavior: a wild sample is treated as an outlier candidate when its removal moves the element-wise median of wild-data gradients toward the InD reference gradient. We establish error bounds showing that this filtering rule controls both inlier removal and outlier retention under the stated contamination, concentration, and separation conditions. Across open-world benchmarks, Medix consistently improves over existing baselines, aligning with the theoretical analysis and demonstrating that median-based filtering is an effective mechanism for learning OOD detectors from unlabeled wild mixtures.

1 Introduction

Machine learning systems deployed in realistic environments must satisfy safety and reliability requirements that closed-world training data cannot fully represent. A central failure mode is induced by out-of-distribution (OOD) inputs: samples originating from categories or sources outside the training label space should be detected and rejected rather than assigned ordinary in-distribution (InD) predictions. Standard neural networks remain susceptible to these inputs and often provide no explicit mechanism for separating them from legitimate InD examples at inference time (Nguyen et al., 2015).

OOD identification is difficult because the relevant unknown distributions are, by definition, absent from supervised InD training. This limitation has motivated methods that incorporate additional “in-the-wild” data into detector learning (Katz-Samuels et al., 2022a; Du et al., 2024a). In particular, Katz-Samuels et al. (2022a) use unlabeled wild data to regularize training while retaining the primary objective of classifying labeled InD examples. This setting is compelling both statistically and operationally: unlabeled wild data can be collected from the environment in which the model is expected to operate, thereby exposing learning to samples that more closely resemble the OOD inputs encountered after deployment.

The same practical advantage creates the central technical obstacle. Wild data is not a clean OOD resource; it is a contaminated mixture of InD and OOD samples, with no membership annotation for individual points. A detector trained from this source must therefore extract reliable outlier evidence from the mixture before the extracted samples can be used as training signal. Despite its importance

for open-world OOD detection, this mixture-filtering problem remains comparatively underdeveloped, and its theory is especially limited: to the best of our knowledge, Du et al. (2024a) is the only prior work that establishes formal guarantees for the unlabeled “in-the-wild” setting.

Recent work on data pruning further shows that median-based criteria can provide robust and informative selection rules (Acharya et al., 2024). These developments motivate the question addressed in this paper:

How can median-based methods leverage unlabeled wild data to facilitate OOD detection with theoretical guarantees?

We answer this question by developing a median-centric formulation of OOD detection from unlabeled mixtures. The resulting framework, Medix, casts outlier extraction as an optimization problem over gradient representations and implements a greedy filtering algorithm that selects candidate OOD samples according to their effect on the element-wise median of wild-data gradients. The analysis proves that InD samples incorrectly flagged as outliers are controlled by two interpretable quantities: a contamination term, which captures the influence of OOD points and remains bounded when the OOD proportion is below 50%, and a concentration term, which uses the sub-Gaussian behavior of InD gradients to control their deviation from the mean InD gradient. Together, these terms give high-probability guarantees for median-based filtering in the unlabeled in-the-wild setting and explain the robustness of the procedure under the mixture configurations covered by the assumptions.

Empirically, we evaluate Medix against two families of baselines: methods trained only with InD data and methods that additionally exploit auxiliary unlabeled data. On CIFAR-100 (Krizhevsky et al., 2009), Medix improves average FPR95 over the strong KNN+ baseline (Sun et al., 2022) by 40.98%. Unlike Outlier Exposure (Hendrycks et al., 2019), which assumes a clean auxiliary dataset separable from InD data, Medix operates directly on the contaminated wild mixture while achieving stronger performance. Relative to WOODS (Katz-Samuels et al., 2022a), Medix reduces average FPR95 by 1.32% on CIFAR-100 and 2.60% on CIFAR-10, demonstrating that explicit median-based filtering improves detector training in the wild-data regime.

The paper makes three connected contributions:

- C1)** We introduce Medix, a median-centric framework that transforms an unlabeled wild mixture into detector supervision by extracting candidate outliers and training a binary OOD detector with the extracted candidates and labeled InD samples.
- C2)** We establish theoretical guarantees for median-based filtering in unlabeled mixtures. The analysis bounds both InD samples incorrectly flagged as outliers and OOD samples retained as inliers through contamination, concentration, and separation effects that characterize when the median rule is reliable.
- C3)** We evaluate Medix across eleven InD-OOD pairs against 20 competitive baselines. The results show consistent gains over all baselines, and the outlier-extraction study reports a low filtering error rate (e.g., 12.5%; see Figure 2), empirically corroborating the predicted behavior of the median filter.

2 Preliminaries and Problem Setup

This section defines the OOD detection setting used throughout the paper, including the labeled InD sample, the unlabeled wild mixture, the classifier and detector parameterizations, and the evaluation objectives.

Labeled In-Distribution Data. Let $\mathcal{X}$ denote the input space and let $\mathcal{Y} = \{1, \dots, K\}$ denote the label space for InD classes. A labeled dataset $\mathcal{S}_{\text{in}} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ contains n independently and identically distributed (i.i.d.) samples from $\mathbb{P}_{XY}$, an unknown joint distribution on $\mathcal{X} \times \mathcal{Y}$. We denote by $\mathbb{P}_{\text{in}}$ the marginal distribution of $\mathbb{P}_{XY}$ on $\mathcal{X}$ and use $\mathcal{S}_{\text{in}}$ to train the InD model.

Out-of-distribution detection. We study the deployment scenario in which a model trained on labeled InD data encounters inputs whose labels fall outside the training label space, i.e., for some label $y \notin \mathcal{Y}$. The detector should accept inputs generated by the InD distribution and reject inputs generated by an OOD source, thereby preventing ordinary class predictions on samples outside the modeled label space.

87 **Unlabeled wild data.** A persistent obstacle in OOD detection is the absence of labeled OOD samples:
 88 possible OOD inputs are numerous, heterogeneous, and costly to annotate. Following Katz-Samuels
 89 et al. (2022a), we therefore incorporate unlabeled wild data, $\mathcal{S}_{\text{wild}} = \{\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_m\}$, into the learning
 90 framework. This wild sample is a deployment-relevant mixture of InD and OOD inputs that can be
 91 collected after training a model on $\mathcal{S}_{\text{in}}$. Following Du et al. (2024a); Katz-Samuels et al. (2022a), we
 92 use the Huber contamination model (Huber, 1964) to characterize the marginal distribution of the
 93 wild data

$$\mathbb{P}_{\text{wild}} := (1 - \pi)\mathbb{P}_{\text{in}} + \pi\mathbb{P}_{\text{out}}, \quad \pi \in (0, 1], \quad (1)$$

94 where π denotes the contamination proportion and $\mathbb{P}_{\text{out}}$ is the OOD distribution over $\mathcal{X}$. The case
 95 $\pi = 0$ contains no OOD samples and therefore reduces to a trivial detection problem.

96 **Models and Loss Functions.** Let $f_\phi : \mathcal{X} \rightarrow \mathbb{R}^K$ denote the InD classifier with parameter $\phi \in \Phi$,
 97 where Φ is the classifier parameter space. The classifier outputs scores over the $K = |\mathcal{Y}|$ InD
 98 classes, and the labeled InD loss is $\ell : \mathbb{R}^K \times \mathcal{Y} \rightarrow \mathbb{R}$. For detection, we introduce a binary
 99 classifier $g_\theta : \mathcal{X} \rightarrow \mathbb{R}$ with parameter $\theta \in \Theta$. Its binary loss is denoted by $\ell_b(g_\theta(x), y_b)$, where
 100 $y_b \in \mathcal{Y}_b := \{y_+, y_-\}$, with $y_+ > 0$ assigned to InD samples and $y_- < 0$ assigned to OOD samples.

101 **Learning objective.** The learning objective uses both the labeled InD distribution $\mathbb{P}_{\text{in}}$ and the
 102 unlabeled wild distribution $\mathbb{P}_{\text{wild}}$ to train the OOD detector g_θ and the multi-class classifier f_ϕ . At
 103 test time, detection and classification performance are evaluated by the following metrics:

$$\begin{aligned} \downarrow \text{FPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{out}}^{\text{test}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{TPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{in}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{Acc}(f_\phi) &= \mathbb{E}_{(\mathbf{x}, y) \sim \mathbb{P}_{\mathcal{X}\mathcal{Y}}} [\mathbb{I}\{f_\phi(\mathbf{x}) = y\}], \end{aligned}$$

104 where $\mathbb{I}\{\cdot\}$ denotes the indicator function, and $\mathbb{P}_{\text{out}}^{\text{test}}$ represents the OOD test data distribution.

105 3 Method: Median-Centric Framework for OOD Detection

106 This section presents **Medix**, a median-centric learning framework for OOD detection with unlabeled
 107 wild data. Instead of relying only on labeled InD samples or assuming access to a clean auxiliary
 108 OOD set, Medix first extracts candidate outliers from the contaminated wild mixture and then uses
 109 those candidates to train a detector. The framework consists of two stages: **1) Outlier Extraction:**
 110 a median-based filtering procedure that isolates candidate OOD samples from unlabeled wild data
 111 (Section 3.1); and **2) Detector Training:** binary detector learning with labeled InD samples and the
 112 extracted outlier candidates (Section 3.2). For the second stage, we follow the protocol introduced by
 113 Du et al. (2024a). This decomposition makes the central mechanism explicit: robust filtering converts
 114 an unlabeled InD–OOD mixture into training evidence for an OOD detector, thereby matching the
 115 open-world setting in which clean OOD labels are unavailable.

116 3.1 Extracting Candidate Outliers from the Wild Data

117 To isolate candidate outliers from the wild mixture $\mathcal{S}_{\text{wild}}$, Medix defines an optimization criterion on
 118 model-parameter gradients. These gradients are computed using a classifier f_ϕ trained only on the
 119 labeled InD sample $\mathcal{S}_{\text{in}}$, so wild-data deviations are measured relative to an InD reference rather than
 120 to labeled OOD examples.

121 **Reference gradient estimation from InD data.** The first step estimates a reference gradient from
 122 the labeled InD data $\mathcal{S}_{\text{in}}$. We train f_ϕ on $\mathcal{S}_{\text{in}}$ by empirical risk minimization (ERM):

$$\phi_{\mathcal{S}_{\text{in}}} \in \arg \min_{\phi \in \Phi} \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi), \quad \text{where } \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi) = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \ell(f_\phi(\mathbf{x}_i), y_i), \quad (2)$$

123 where $\phi_{\mathcal{S}_{\text{in}}}$ denotes the learned parameters. After training, we compute the mean InD gradient $\bar{\nabla}_{\text{in}}$ by
 124 averaging the loss gradients, taken with respect to the model parameters, over the labeled InD sample:

$$\bar{\nabla}_{\text{in}} = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}(\mathbf{x}_i), y_i). \quad (3)$$

The vector $\bar{\nabla}_{\text{in}}$ is the reference against which Medix compares the aggregate gradient behavior of wild-data subsets.

Motivation. The filtering criterion is motivated by the empirical observation that increasing OOD contamination in $\mathcal{S}_{\text{wild}}$ moves the median wild-gradient profile away from the average InD gradient $\bar{\nabla}_{\text{in}}$. We examine this behavior using CIFAR-10 (Krizhevsky et al., 2009) as the InD dataset and SVHN (Netzer et al., 2011) as the OOD dataset. The initial $\mathcal{S}_{\text{wild}}$ contains 10,000 CIFAR-10 samples disjoint from the training set used to obtain $\phi_{\mathcal{S}_{\text{in}}}$ and compute $\bar{\nabla}_{\text{in}}$. We then incrementally add SVHN samples and measure the L_2 deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median (EWM) of wild-data gradients:

$$\left\| \bar{\nabla}_{\text{in}} - \text{EWM} \left(\left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}_{\text{wild}}} \right) \right\|.$$

Figure 1 shows a clear monotonic increase in this deviation as OOD samples are added. This pattern directly motivates the Medix filtering rule: removing genuine OOD points should reduce the median-gradient deviation toward the InD reference, and filtering can stop once further removals no longer produce a meaningful decrease, as determined by the threshold ϵ .

Filtering potential outliers from unlabeled wild data. Motivated by Figure 1, we formulate outlier extraction as an optimization problem over subsets of the wild sample, with the objective of identifying $\mathcal{S}_{\text{out}}^*$ within $\mathcal{S}_{\text{wild}}$:

$$\mathcal{S}_{\text{in}}^* = \arg \min_{\mathcal{S} \subseteq \mathcal{S}_{\text{wild}}} \left\| \bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}}) \right\|, \text{ where } G_{\mathcal{S}} = \left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}}. \quad (4)$$

Here $\text{EWM}(\cdot)$ denotes the element-wise median function, and $\hat{y}_{\tilde{\mathbf{x}}_i}$ is the predicted label for a wild sample $\tilde{\mathbf{x}}_i$. For notational simplicity, we write $G_{\mathcal{S}} = \{\nabla \ell(\tilde{\mathbf{x}}_i)\}_{i \in \mathcal{S}}$. The optimization seeks the subset $\mathcal{S}$ of $\mathcal{S}_{\text{wild}}$ whose EWM gradient is closest to the InD reference gradient $\bar{\nabla}_{\text{in}}$. In light of Figure 1, the minimizing subset $\mathcal{S}_{\text{in}}^*$ is interpreted as the InD component of the wild mixture, and its complement $\mathcal{S}_{\text{out}}^* = \mathcal{S}_{\text{wild}} \setminus \mathcal{S}_{\text{in}}^*$ provides the corresponding candidate OOD set.

Directly solving (4) is computationally prohibitive for large $\mathcal{S}_{\text{wild}}$. Medix therefore uses the greedy leave-one-out approximation in Algorithm 1, iteratively removing the samples whose exclusion most reduces the median-gradient deviation from the InD reference.

At iteration t , the algorithm computes the EWM of gradients for the remaining wild subset $\mathcal{S}$ and measures its distance from $\bar{\nabla}_{\text{in}}$ as $d_t = \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$. For each candidate sample $i \in \mathcal{S}$, Medix recomputes the EWM after removing that sample, $\text{EWM}(G_{\mathcal{S} \setminus \{i\}})$, and evaluates the induced decrease $\delta_i = d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$. The k samples producing the largest decreases are added to the outlier set and removed from $\mathcal{S}$. The procedure terminates when the largest remaining decrease falls below ϵ or the maximum number of iterations is reached. This stopping rule reflects the monotonic trend in Figure 1: after the principal OOD mass has been removed, additional removals produce only small changes in L_2 distance and increasingly risk discarding InD samples.

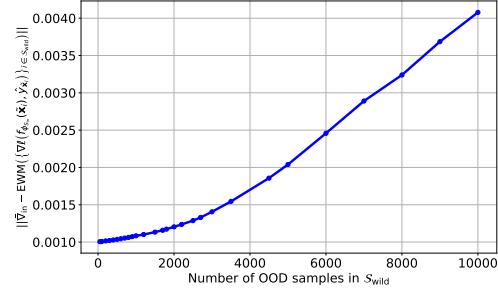


Figure 1: Median-gradient distance from the InD reference increases as OOD samples are added to $\mathcal{S}_{\text{wild}}$.

3.2 Training the OOD Detector with Candidate Outliers

After Medix identifies the candidate outlier set $\hat{\mathcal{S}}_{\text{out}}$ from wild data, we train the OOD detector g_{θ} to separate labeled InD samples from the extracted candidates, following the protocol of Du et al. (2024a). The objective imposes separability at the zero decision threshold by assigning positive outputs to labeled InD samples $\mathbf{x} \in \mathcal{S}_{\text{in}}$ and negative outputs to candidate outliers $\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}$. The resulting loss is

$$\begin{aligned} \mathcal{L}_{\mathcal{S}_{\text{in}}, \hat{\mathcal{S}}_{\text{out}}}(g_{\theta}) &= \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_{\theta}) + \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_{\theta}), \\ \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_{\theta}) &= \mathbb{E}_{\mathbf{x} \in \mathcal{S}_{\text{in}}} \mathbb{I}\{g_{\theta}(\mathbf{x}) \leq 0\}, \\ \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_{\theta}) &= \mathbb{E}_{\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}} \mathbb{I}\{g_{\theta}(\tilde{\mathbf{x}}) > 0\}. \end{aligned} \quad (5)$$

Algorithm 1 Iterative Outlier Detection via **Medix**

Require: $\bar{\nabla}_{\text{in}}, G_i = \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i})}$, maximum iterations T , hyperparameters ϵ, k

- 1: Initialize $\mathcal{S} \leftarrow \mathcal{S}_{\text{wild}}$ (wild set), $\mathcal{O} \leftarrow \emptyset$ (outliers), $t \leftarrow 0$ (iteration), $d_t \leftarrow 0$, $\delta_{\text{max}} \leftarrow \infty$ (deviations), $\mathcal{I}_k \leftarrow \emptyset$
- 2: **while** $t \leq T$ or $|\delta_{\text{max}}| > \epsilon$ **do**
- 3: $\mathcal{O} \leftarrow \mathcal{O} \cup \{\tilde{\mathbf{x}}_i : i \in \mathcal{I}_k\}$ ▷ Add outliers to set $\mathcal{O}$
- 4: $d_t \leftarrow \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$ ▷ Compute L2 deviation
- 5: **for each** $i \in \mathcal{S}$ **do**
- 6: $\delta_i \leftarrow d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$
- 7: **end for**
- 8: $\mathcal{I}_k \leftarrow \text{indices}(\text{top-}k(\{\delta_i\}_{i \in \mathcal{S}}))$ ▷ Select top- k indices
- 9: $\mathcal{S} \leftarrow \mathcal{S} \setminus \mathcal{I}_k$ ▷ Remove outliers from $\mathcal{S}$
- 10: $\delta_{\text{max}} \leftarrow \max_{i \in \mathcal{S}} \{\delta_i\}$, $t \leftarrow t + 1$
- 11: **end while**
- 12: **return** $\mathcal{S}_{\text{out}} = \mathcal{O}$ ▷ Return detected outliers

Because the 0/1 loss is non-differentiable, we optimize a differentiable sigmoid-based binary surrogate that preserves the intended decision rule while making training tractable. The detector objective is combined with the InD risk in Equation 2, so detector training uses the extracted outlier signal without discarding the classifier’s InD predictive objective. This joint optimization yields the detector evaluated on OOD samples drawn from $\mathbb{P}_{\text{out}}$.

4 Theoretical Analysis

We now formalize the behavior of Medix’s filtering stage. The following results upper-bound the two error modes induced by filtering: InD points incorrectly removed as outliers and OOD points incorrectly retained as inliers. These bounds make explicit how EWM filtering depends on contamination, gradient concentration, and, for OOD retention, separation from the InD gradient profile. Detailed proofs appear in Appendix C.

Theorem 4.1 (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$, and fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the EWM filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

Theorem 4.1 decomposes inlier filtering error into a *concentration term*, which decreases with sample size and accounts for finite-sample fluctuations of InD gradients, and a *contamination term*, which measures the influence of OOD gradients on the EWM. The bound remains controlled when the contamination ratio satisfies $\pi < 0.5$, matching the median’s robustness to minority contamination. The complementary question is how many true outliers are not removed; the next theorem bounds this OOD misclassification rate.

Theorem 4.2 (Outlier Misclassification Bound). *Assume that the gradients of OOD points in*

$\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ_{out}^2 . Suppose the mean OOD gradient μ_{out} satisfies a separation condition:

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d},$$

for some $\Delta > 0$. Then, for any tolerance $\epsilon \in (0, \Delta)$ and confidence level $\delta \in (0, 1)$, with probability at least $1 - \delta$, the outlier misclassification rate satisfies:

$$\text{ERR}_{\text{out}} \leq \underbrace{2d \exp\left(-\frac{(\Delta - \epsilon)^2}{2\sigma_{\text{out}}^2}\right)}_{\text{Separation term}} + \underbrace{\sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}}_{\text{Concentration term}} + \underbrace{\frac{1 - \pi}{2\pi}}_{\text{Contamination term}}.$$

188

189 Theorem 4.2 complements the inlier result by bounding the fraction of OOD points mistakenly
 190 retained as InD. Together, Theorems 4.1 and 4.2 provide a two-sided robustness guarantee for Medix
 191 filtering. The two error rates are governed by three effects:

192 **Contamination Effect.** In Theorem 4.1, the median criterion remains stable under a nontrivial OOD
 193 fraction π provided $\pi < 0.5$. The term $\pi/[2(1 - \pi)]$ increases with π and captures the extent to
 194 which OOD gradients can shift the EWM. Conversely, Theorem 4.2 includes the reverse penalty $\frac{1-\pi}{2\pi}$,
 195 reflecting the difficulty of isolating OOD points when they are underrepresented in the mixture.

196 **Concentration Effect.** Both bounds use the sub-Gaussian behavior of gradient coordinates. For InD
 197 data, concentration near $\bar{\nabla}_{\text{in}}$ stabilizes the median under finite samples. For OOD data, concentration
 198 around μ_{out} quantifies the probability that an OOD gradient remains close enough to the InD profile
 199 to evade filtering. These terms decay at the stated $1/\sqrt{m}$ rate with the relevant sample size.

200 **Separation Effect.** Theorem 4.2 additionally requires separation between the OOD mean gradient and
 201 the InD reference gradient. The exponential term $\exp(-(\Delta - \epsilon)^2/2\sigma_{\text{out}}^2)$ expresses this dependence:
 202 larger separation makes it less likely that OOD gradients are retained by the median filter.

Theorems 4.1 and 4.2 jointly establish that, with high probability, median filtering controls both directions of error in unlabeled InD–OOD mixtures. InD removal is bounded by contamination and concentration effects, whereas OOD retention is controlled by separation, concentration, and reverse contamination. These results give Medix a principled basis for extracting outlier candidates before detector training.

203

204 *Remark 4.3. Sub-Gaussian Assumption for Gradient Coordinates.* The coordinate-wise sub-Gaussian
 205 assumption on InD gradients enables the concentration bounds used in the analysis. Figure 4a
 206 provides empirical support for this assumption: the histogram of InD gradient values is concentrated
 207 around the mean and exhibits light tails, which is consistent with sub-Gaussian behavior. Figure 4b
 208 further compares empirical gradient quantiles with Gaussian theoretical quantiles in a Q-Q plot (Wilk
 209 and Gnanadesikan, 1968); the alignment with the 45-degree reference line indicates that the empirical
 210 gradient distribution is close to Gaussian, and Gaussian variables form a subclass of sub-Gaussian
 211 variables. *We also provide a looser version of these bounds, which **removes the sub-Gaussian***
 212 ***assumption** and holds under merely bounded second moments. This version is presented in Theorem*
 213 *C.3 (Appendix C.3). While the rates degrade, the core robustness guarantee of Medix still holds in*
 214 *this broader setting.*

215 5 Experiments

216 This section evaluates Medix across multiple InD–OOD dataset pairs and benchmarks it against
 217 20 widely used baselines. All experiments are performed on hardware equipped with NVIDIA
 218 A100-SXM4-80GB GPUs. The code required to reproduce the reported results will be released upon
 219 acceptance.

220 5.1 Models, Datasets, and Baselines

221 **Datasets.** We follow the experimental protocol of Katz-Samuels et al. (2022a), which introduced
 222 learning OOD detectors with wild data, so that comparisons to prior work are direct. CIFAR-10

and CIFAR-100 serve as the InD datasets ($\mathbb{P}_{\text{in}}$). For OOD testing, we use a suite of natural-image datasets—PLACES365 (Zhou et al., 2017), SVHN, TEXTURES (Cimpoi et al., 2014), and LSUN-RESIZE & LSUN-C (Yu et al., 2015)—as OOD datasets ($\mathbb{P}_{\text{out}}$). To simulate wild data ($\mathbb{P}_{\text{wild}}$), we combine a subset of InD data ($\mathbb{P}_{\text{in}}$) with OOD data ($\mathbb{P}_{\text{out}}$) using the default mixing parameter $\pi = 0.5$. For example, when PLACES365 is the OOD test set, the wild mixture is constructed by combining CIFAR samples with PLACES365 samples, and PLACES365 is then used for OOD evaluation. We repeat this construction across all OOD datasets and baselines. Each InD CIFAR dataset is split into two halves: the first 25,000 images train $\phi_{\mathcal{S}_{\text{in}}}$, and the remaining images are used to construct the wild mixture $\mathcal{S}_{\text{wild}}$. For gradient computation, we use the penultimate-layer weights because this representation has been shown to be particularly informative for OOD detection (Huang et al., 2021a).

Evaluation metrics. We report three standard metrics: (1) False Positive Rate (FPR95 $\downarrow$) of OOD samples when the True Positive Rate of InD samples is fixed at 95%, (2) Area Under the Receiver Operating Characteristic Curve (AUROC $\uparrow$), and (3) InD Classification Accuracy (InD Acc $\uparrow$).

Baselines. The comparison includes competitive baselines grouped by their training information: methods trained only on InD data and methods trained with both InD and wild data. Among methods trained exclusively on InD data ($\mathbb{P}_{\text{in}}$), we compare Medix with established OOD detection scores and post hoc methods, including Maximum Softmax Probability (MSP) (Hendrycks and Gimpel, 2016), ODIN (Liang et al., 2018), Mahalanobis Distance (Lee et al., 2018a), Energy Score (Liu et al., 2020b), ReAct (Sun et al., 2021), DICE (Sun and Li, 2022), KNN Distance (Sun et al., 2022), and ASH (Djurisic et al., 2022); these methods use a model trained with softmax cross-entropy loss. We also include contrastive-loss methods, CSI (Tack et al., 2020a) and KNN+ (Sun et al., 2022), to broaden the comparison. For methods that leverage both InD and wild data, we compare against Outlier Exposure (OE) (Hendrycks et al., 2019) and energy-regularization learning (Liu et al., 2020b), which regularize training by promoting lower confidence or higher energy on outlier data. We also compare with WOODS (Katz-Samuels et al., 2022a), which uses unlabeled wild data for OOD detection through constrained optimization. Finally, we include recent baselines, including CONJ (Peng et al., 2024) and DRL (Zhang et al., 2024), to make the evaluation current and comprehensive.

5.2 Experimental Setup

Following WOODS (Katz-Samuels et al., 2022a), we use a Wide ResNet architecture (Zagoruyko, 2016) with 40 layers and width factor 2 for the InD classifier $\phi_{\mathcal{S}_{\text{in}}}$. The classifier is trained by stochastic gradient descent with momentum 0.9, weight decay 0.0005, and initial learning rate 0.1. Training runs for 100 epochs with cosine learning-rate decay, batch size 128, and dropout rate 0.3. The Medix hyperparameters ϵ and k are selected from $\{5\text{e-}5, 5\text{e-}4, 5\text{e-}3, 5\text{e-}2\}$ and $\{4\text{k}, 7\text{k}, 10\text{k}, 20\text{k}\}$, respectively, accounting for dataset size and optimizing OOD performance. For the OOD classifier g_θ , we initialize from the pre-trained InD classifier $\phi_{\mathcal{S}_{\text{in}}}$ and add a linear binary-classification layer on the penultimate-layer features. The detector learning rate is 0.001, and fine-tuning proceeds for 100 epochs according to Equation (5). We combine the binary classification loss with the InD classification loss, assign weight 10 to the binary component, and otherwise retain the training parameters used for $\phi_{\mathcal{S}_{\text{in}}}$.

5.3 Results

Table 2 reports the main CIFAR-100 results and shows that Medix outperforms all evaluated OOD detection baselines. Two empirical patterns are especially salient. First, methods trained with both InD and wild data substantially outperform methods trained exclusively on InD data, confirming the value of deployment-relevant unlabeled samples. Relative to KNN+, Medix reduces FPR95 by 52.31% on PLACES365 and 38.24% on TEXTURES. Second, Medix further improves over methods that already use wild data ($\mathbb{P}_{\text{wild}}$): on CIFAR-100, it achieves an average FPR95 of 5.42%, a 1.32% improvement over WOODS, while maintaining competitive InD accuracy of 73.33%. This accuracy difference is consistent with the data split, since Medix is trained on 25,000 labeled InD samples whereas baselines that do not use wild data train on the full CIFAR-100 training set of 50,000 samples. Table 1 gives the corresponding CIFAR-10 results, where Medix again surpasses all baselines and improves FPR95 over WOODS by 7.21% on PLACES365 and 5.25% on TEXTURES.

Table 1: OOD detection performance comparison of Medix and baselines on CIFAR-10 as InD data. Performance averaged over five runs; best results are highlighted in **bold**.

Methods	OOD Datasets												InD ACC $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
Using $\mathbb{P}_m$ only													
MSP	48.49	91.89	59.48	88.20	30.80	95.65	52.15	91.37	59.28	88.50	50.04	91.12	94.84
ODIN	33.35	91.96	57.40	84.49	15.52	97.04	26.62	94.57	49.12	84.97	36.40	90.61	94.84
Mahalanobis	12.89	97.62	68.57	84.61	39.22	94.15	42.62	93.23	15.00	97.33	35.66	93.34	94.84
Energy	35.59	90.96	40.14	89.89	8.26	98.35	27.58	94.24	52.79	85.22	32.87	91.73	94.84
KNN	24.53	95.96	25.29	95.69	25.55	95.26	27.57	94.71	50.90	89.14	30.77	94.15	94.84
ReAct	89.57	81.44	90.44	90.44	14.38	97.21	33.63	93.58	53.63	86.59	36.77	91.48	94.84
DICE	35.44	89.65	46.83	86.69	6.32	98.68	28.93	93.56	53.62	82.20	34.23	90.16	94.84
ASH	6.51	98.65	48.45	88.34	0.90	99.73	4.96	98.92	24.34	95.09	17.03	96.15	94.84
CSI	17.30	97.40	34.95	93.64	1.95	99.55	12.15	98.01	20.45	95.93	17.36	96.91	94.17
KNN+	2.99	99.41	24.69	94.84	2.95	99.39	11.22	97.98	9.65	98.37	10.30	97.99	93.19
Using $\mathbb{P}_m$ and $\mathbb{P}_{wild}$													
OE	1.13	99.53	19.48	94.88	1.91	98.16	0.54	98.84	7.75	98.56	6.16	97.99	94.12
Energy (w/ OE)	5.24	98.72	14.66	96.18	2.35	99.30	4.85	98.62	10.51	97.10	7.52	97.98	94.24
WOODS	0.17	99.91	10.19	98.05	0.31	99.14	0.11	99.38	6.21	98.13	3.40	98.92	94.74
Medix	0.06	99.98	2.98	99.10	0.01	99.98	0.01	99.98	0.96	99.66	0.80	99.74	93.58
(Ours)	± 0.01	± 0.01	± 0.29	± 0.09	± 0.01	± 0.00	± 0.01	± 0.01	± 0.13	± 0.06	± 0.09	± 0.03	± 0.64

Table 2: OOD detection performance comparison of Medix and baselines on CIFAR-100 as InD data. Performance averaged over five runs; best results are highlighted in **bold**.

Methods	OOD Datasets												InD ACC $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
Using $\mathbb{P}_m$ only													
MSP	84.59	71.44	82.84	73.78	66.54	83.79	82.42	75.38	83.29	73.34	79.94	75.55	75.96
ODIN	84.66	67.26	87.88	71.63	55.55	87.73	71.96	81.82	79.27	73.45	75.86	76.38	75.96
Mahalanobis	57.52	86.01	88.83	67.87	91.18	69.69	21.23	96.00	39.39	90.57	59.63	82.03	75.96
Energy	85.82	73.99	80.56	75.44	35.32	93.53	79.47	79.23	79.41	76.28	72.12	79.69	75.96
KNN	66.38	83.76	79.17	71.91	70.96	83.71	77.83	78.85	88.00	67.19	76.47	77.08	75.96
ReAct	74.33	88.04	81.33	74.32	39.30	91.19	79.86	73.69	67.38	82.80	68.44	82.01	75.96
DICE	88.35	72.58	81.61	75.07	26.77	94.74	80.21	78.50	76.29	76.07	70.65	79.39	75.96
ASH	21.36	94.28	68.37	71.22	15.27	95.65	68.18	85.42	40.87	92.29	42.81	87.77	75.96
CSI	64.70	84.97	82.25	73.63	38.10	92.52	91.55	63.42	74.70	92.66	70.26	81.44	69.90
KNN+	32.21	93.74	68.30	75.31	40.37	86.13	44.86	88.88	46.26	87.40	46.40	86.29	73.78
Using $\mathbb{P}_m$ and $\mathbb{P}_{wild}$													
OE	2.86	99.05	40.21	88.75	4.13	99.05	1.25	99.38	22.86	94.63	14.26	96.17	73.38
Energy (w/ OE)	2.71	99.34	34.82	90.05	3.27	99.18	2.54	99.23	30.16	94.76	14.70	96.51	72.76
WOODS	0.17	99.80	21.87	93.73	0.48	99.61	1.24	99.54	9.95	95.97	6.74	97.73	73.91
Medix	0.16	99.96	15.99	95.23	0.13	99.98	0.83	99.83	8.02	97.79	5.42	98.96	73.33
(Ours)	± 0.02	± 0.00	± 0.66	± 0.14	± 0.06	± 0.02	± 0.36	± 0.06	± 0.75	± 0.30	± 0.37	± 0.10	± 0.83

A representative visual example of Medix. We further examine Algorithm 1 by visualizing outlier extraction from wild data $\mathcal{S}_{wild}$ in a two-dimensional synthetic setting. The simplified construction makes the filtering behavior directly interpretable. InD data are sampled from three multivariate Gaussian distributions representing three classes, with mean vectors $[-2, 0]$, $[2, 0]$, and $[0, 2\sqrt{3}]$, respectively, and common covariance $0.25 \cdot I$. OOD data are sampled from $\mathcal{N}([20, 2\sqrt{3}], 0.25 \cdot I)$. Figure 2 shows that the outliers identified by Medix largely coincide with the ground truth: the algorithm flags 87.5% of actual OOD samples as outliers, yielding the reported 12.5% extraction error rate.

Additional studies and insights. Additional studies in Appendix A support the main empirical conclusions from complementary perspectives. They include hyperparameter selection and sensitivity analysis (Appendix A.2), which shows that Medix remains robust across choices of ϵ and k ; a comparison between EWM and geometric median (Appendix A.1), which shows that EWM is more sensitive to distributional shifts in this filtering procedure; a comprehensive comparison with methods using competitive contrastive learning objectives (Appendix A.3), where Medix continues to outperform the evaluated alternatives; an unseen-OOD evaluation with $\mathbb{P}_{out}^{test} \neq \mathbb{P}_{out}$ (Appendix A.4); profiling of computation and memory efficiency (Appendix A.6); an assessment of pseudo-label quality (Appendix A.5); and a comparison with semi-supervised open-set recognition methods (Appendix A.7).

6 Related Work

OOD detection has become an increasingly active area of research (Fort et al., 2021; Yang et al., 2024; Fang et al., 2022; Zhu et al., 2022; Yang et al., 2022; Wang et al., 2022c; Galil et al., 2023; Djuricic et al., 2023; Tao et al., 2023; Zheng et al., 2023; Wang et al., 2022b, 2023b; Uppaal et al., 2023; Zhu et al., 2023; Bai et al., 2023; Ming and Li, 2024; Zhang et al., 2023; Ghosal et al., 2024; Abbas et al., 2025). A major line of work detects OOD data through scoring functions that assess distributional deviation, including distance-based methods (Lee et al., 2018b; Tack et al., 2020b; Ren

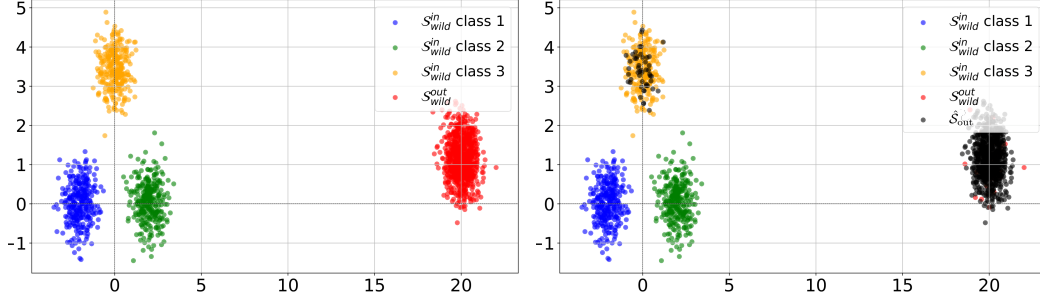


Figure 2: Medix applied to unlabeled wild data. (a) Synthetic wild-data construction with InD data S_{wild}^{in} sampled from three multivariate Gaussian distributions and OOD data S_{wild}^{out} added to the mixture. (b) Outliers $\hat{S}_{out}$ filtered by Medix (black), where the extracted set contains InD data S_{wild}^{in} at an error rate of only 12.5%.

et al., 2021; Sehwan et al., 2021; Sun et al., 2022; Du et al., 2022a; Ming et al., 2023; Ren et al., 2022), gradient-based score (Huang et al., 2021b), energy-based score (Liu et al., 2020c; Wang et al., 2021; Wu et al., 2023), confidence-based approaches (Bendale and Boulton, 2016; Hendrycks and Gimpel, 2017; Liang et al., 2018), and Bayesian methods (Gal and Ghahramani, 2016; Lakshminarayanan et al., 2017; Maddox et al., 2019; Malinin and Gales, 2019; Wen et al., 2020; Kristiadi et al., 2020).

A complementary line of work incorporates OOD-oriented regularization during training (Malinin and Gales, 2018; Geifman and El-Yaniv, 2019; Hein et al., 2019; Meinke and Hein, 2020; Jeong and Kim, 2020; Liu et al., 2020a; Van Amersfoort et al., 2020; Yang et al., 2021; Wei et al., 2022; Du et al., 2022b, 2023; Wang et al., 2023a). For example, such regularization can reduce model confidence (Lee et al., 2017; Hendrycks et al., 2019) or increase its energy (Liu et al., 2020c; Du et al., 2022c; Ming et al., 2022) on the OOD data. Most of these regularization methods assume the availability of an auxiliary OOD dataset.

Several studies (Zhou et al., 2021; Katz-Samuels et al., 2022b; He et al., 2023) relax the purely labeled-data setting by incorporating unlabeled wild data (Katz-Samuels et al., 2022a; Geng et al., 2025), but they do not provide the same median-based outlier extraction mechanism studied here. Du et al. (2024a,b) introduce explicit outlier filtering, yet their thresholding approach differs from Medix’s median-centric criterion. In addition, Katz-Samuels et al. (2022a); Du et al. (2024a) assume batch-level mixing, where each batch has a prescribed InD/OOD ratio; in large outsourced datasets, by contrast, such structured mixing is typically unavailable because samples are mixed at the dataset level. Medix addresses this setting without relying on batch-level structure. Positive-unlabeled learning also trains classifiers from positive and/or unlabeled data (Letouzey et al., 2000; Hsieh et al., 2015; Plessis et al., 2015; Niu et al., 2016; Gong et al., 2018; Chapel et al., 2020; Garg et al., 2021; Xu and Denil, 2021; Garg et al., 2022; Zhao et al., 2022; Acharya et al., 2022). The distinction is that these methods primarily separate $\mathbb{P}_{out}$ from $\mathbb{P}_{in}$, whereas Medix identifies outliers in unlabeled wild data and then uses them to train a dedicated OOD detector with provably bounded filtering error under the stated assumptions.

7 Conclusions

This work introduced Medix, a median-centric framework for OOD detection from unlabeled in-the-wild data. By using the robustness of the median in gradient space, Medix filters candidate outliers from contaminated unlabeled mixtures and converts them into supervision for a binary OOD detector. The theoretical analysis bounds inlier misclassification under contamination up to 50%, with error controlled by sub-Gaussian concentration and contamination effects, and gives a complementary OOD-retention bound under the stated separation condition. Across diverse benchmarks, Medix improves over 20 baselines: it reduces average FPR95 by 40.98% relative to strong baselines such as KNN+ and outperforms methods including WOODS and DRL, while the synthetic outlier-extraction study reports an error rate as low as 12.5%. Together, the theory and experiments establish median-based filtering as a principled and effective route for training OOD detectors from unlabeled wild mixtures.

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A Additional Studies

A.1 Element-wise Median vs Geometric Median

A natural design question is why Medix uses the element-wise median rather than the geometric median (Acharya et al., 2024) for outlier extraction. We answer this question with a preliminary experiment using CIFAR-10 as the InD dataset and SVHN as the OOD dataset. The wild sample $\mathcal{S}_{\text{wild}}$ contains 5k CIFAR-10 samples that are disjoint from the training set used to train $\phi_{\mathcal{S}_{\text{in}}}$, which is used to compute $\bar{\nabla}_{\text{in}}$. We then incrementally add SVHN OOD samples to $\mathcal{S}_{\text{wild}}$ and track both the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median of wild-data gradients and the proportion of OOD samples removed. Figure 3 shows that, for the same number of OOD samples in the wild dataset, the element-wise median identifies a substantially larger proportion of OOD samples than the geometric median. This result supports the use of the element-wise median as the filtering statistic in Medix.

Table 3: Effect of hyperparameters ϵ and k on OOD detection.

Method	FPR95↓	ϵ	k
DICE	88.35	—	—
ASH	21.36	—	—
CSI	64.70	—	—
KNN+	32.21	—	—
OE	2.86	—	—
Energy	2.71	—	—
Medix	0.16	0.005	20000
Medix	0.20	0.0005	20000
Medix	0.68	0.005	10000

A.2 Hyperparameter Selection and Sensitivity Analysis

We assess the sensitivity of Medix to the filtering hyperparameters ϵ and k using CIFAR-100 as the InD dataset and SVHN as the OOD dataset. Table 3 shows that Medix remains robust across the evaluated settings. It achieves an FPR95 of 0.16 with $\epsilon = 0.005$ and $k = 20000$; when ϵ is reduced to 0.0005 or k is halved to 10000, performance remains strong, with FPR95 values of 0.20 and 0.68, respectively. These results indicate graceful degradation rather than sensitivity to a single hyperparameter configuration, while still surpassing the listed baselines. This robustness is important for deployment settings in which exhaustive tuning is impractical.

A.3 Comprehensive comparison with recent competitive methods

We extend the comparison to competitive baselines that use diverse contrastive learning objectives, including SSD+ (Sehwag et al., 2021), ProxyAnchor (Kim et al., 2020), and CIDER Ming et al. (2023). Although Medix does not use a contrastive objective, including these methods tests whether median-based filtering remains competitive against representation-learning approaches. We also include recent methods such as CONJ (Peng et al., 2024), DRL (Zhang et al., 2024), Vim (Wang et al., 2022a), and VOS (Du et al., 2022c). For this experiment, CIFAR-100 is the InD dataset and the ResNet-34 model¹ is trained following the setup in Ming et al. (2023); Zhang et al. (2024). The model is trained by stochastic gradient descent with momentum 0.9 and weight decay 10^{-4} for 500 epochs, using initial learning rate 0.5 with cosine scheduling and batch size 512. We use the

¹Since ResNet-34 is the standard model across these works, we adopt the same model to ensure consistency and facilitate a fair comparison.

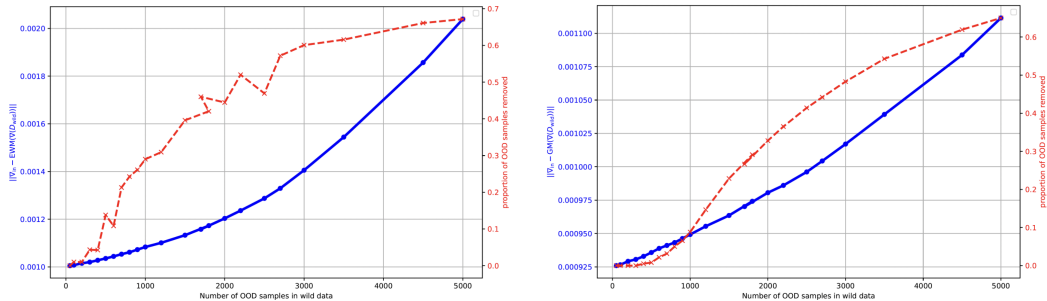


Figure 3: Comparison of element-wise median (EWM) and geometric median (GM).

Table 4: OOD detection performance comparison of Medix and recent competitive baselines on CIFAR-100 as InD data. Performance averaged over five runs; best results are highlighted in **bold**.

Methods	OOD Datasets									
	SVHN		PLACES365		LSUN-C		TEXTURES		AVERAGE	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MSP	78.89	79.80	84.38	74.21	83.47	75.28	86.51	72.53	83.31	75.46
Mahalanobis	87.09	80.62	84.63	73.89	84.15	79.43	61.72	84.87	79.40	79.70
ODIN	70.16	84.88	82.16	75.19	76.36	80.10	85.28	75.23	78.49	78.85
Energy	66.91	85.25	81.41	76.37	59.77	86.69	79.01	79.96	71.78	82.07
ReAct	50.93	88.75	83.55	73.10	64.02	80.31	64.40	81.95	65.72	81.03
KNN	46.25	90.39	82.08	75.44	60.85	85.61	62.39	83.95	62.89	83.85
Vim	73.42	84.62	85.34	69.34	86.96	69.74	74.56	76.23	80.07	74.98
VOS	43.24	82.80	76.85	78.63	73.61	84.69	57.57	87.31	62.82	83.36
CSI	44.53	92.65	79.08	76.27	75.58	83.78	61.61	86.47	65.20	84.79
ProxyAnchor	87.21	82.43	70.10	79.84	37.19	91.68	65.64	84.99	65.04	84.74
SSD+	31.19	94.19	77.74	79.90	79.39	85.18	66.63	86.18	63.74	86.36
KNN+	39.23	92.78	80.74	77.58	48.99	89.30	57.15	88.35	56.53	87.00
ASH	52.96	90.19	72.62	76.38	75.18	76.52	56.17	86.75	64.23	82.46
CIDER	23.09	95.16	79.63	73.43	16.16	96.33	43.87	90.42	40.69	88.84
CONJ	46.19	90.44	80.81	75.83	60.45	85.90	62.13	83.77	62.40	83.99
DRL	20.15	94.07	76.64	77.55	16.97	94.63	31.97	92.09	36.43	89.59
Medix	0.48	99.87	24.52	93.42	0.73	99.84	8.99	97.92	8.68	97.76
(Ours)	± 0.05	± 0.02	± 1.76	± 0.36	± 0.08	± 0.03	± 0.72	± 0.24	± 0.65	± 0.16

checkpoints provided by Ming et al. (2023)². Table 4 shows that Medix outperforms these competitive baselines, including those trained with contrastive learning objectives, achieving an average FPR95 of 8.68% and an average AUROC of 97.76%. Medix also provides explicit filtering-error guarantees, a theoretical property not supplied by most alternatives in this comparison.

A.4 Evaluation on Large-Scale, Complex Unseen OOD Settings

We next evaluate Medix in a large-scale unseen-OOD setting. This setting is challenging because it uses a large wild OOD dataset ($\mathbb{P}_{\text{out}}$), increasing both data complexity and computational demand, and because the test-time OOD distribution ($\mathbb{P}_{\text{out}}^{\text{test}}$) is intentionally different from the wild OOD data used during training, i.e., $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$. This protocol tests whether the extracted wild-data signal remains useful when training-time and test-time OOD sources differ. We use CIFAR-100 as the labeled InD data, the 300K Random Images dataset Hendrycks et al. (2019) as the unlabeled wild OOD data, and SVHN as the test OOD dataset. The 300K Random Images source differs substantially from SVHN, making the setting more demanding than matched wild/test OOD evaluation. Against baselines that also leverage wild data during training, Table 5 shows that Medix obtains the strongest performance, with FPR95 41.29 ± 1.2 and AUROC 87.25 ± 0.9 .

Table 5: Comparison of OOD detection performance on large-scale unseen OOD data.

Method	FPR ↓	AUROC ↑
OE	68.80 ± 2.8	82.89 ± 1.1
Energy (w/ OE)	69.81 ± 2.4	85.59 ± 1.0
WOODS	69.41 ± 2.7	86.76 ± 0.8
Medix (ours)	41.29 ± 1.2	87.25 ± 0.9

A.5 Evaluating the Impact of Pseudo-Label Quality on OOD Filtering and Robustness

We also examine whether pseudo-label quality, measured through the softmax confidence of $\hat{y}_{\tilde{x}_i}$, affects OOD filtering, and whether removing low-confidence pseudo-labels improves robustness. The experiment uses CIFAR-10 as the InD dataset and LSUN-Resize as the OOD dataset. We compute softmax probabilities for each pseudo-labeled sample and discard predictions below a confidence threshold of 0.6, which removes 15.98% of samples from the training set. Performance is essentially unchanged: both the filtered and unfiltered variants achieve FPR95 of 0.01%, while AUROC increases only from 99.98% to 99.99% with filtering. This result indicates that Medix is resilient to noisy or low-confidence pseudo-labels in this setting, and that an additional confidence-based pre-filtering stage provides little practical benefit.

²<https://github.com>

A.6 Evaluating Computation and Memory Efficiency in Medix Filtering

We finally profile whether the Medix filtering phase is computationally feasible and memory-efficient on larger unlabeled pools. The profiling experiments are conducted on an NVIDIA A100-SXM4-80GB GPU with approximately 15,000 unlabeled samples for each InD–OOD pair, using CIFAR-10 and CIFAR-100 as InD datasets and LSUN-Resize as the OOD dataset. Table 6 shows modest GPU memory usage and tractable wall-clock time for the filtering phase, indicating that Medix filtering does not impose excessive memory overhead at this scale.

Table 6: Profiling results for Medix filtering on NVIDIA A100 80GB GPU.

InD–OOD	Wall Clock Time (s)	Peak GPU Memory (MB)	Current GPU Memory (MB)
CIFAR10 – LSUN-Resize	4497.17	99.46	31.74
CIFAR100 – LSUN-Resize	5478.36	99.56	31.79

A.7 Comparison with Semi-Supervised Open-Set Recognition Methods

Medix differs from recent semi-supervised open-set recognition methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) in objective, use of outliers, and technical mechanism. In OpenML Saito et al. (2021), for example, the central problem is handling outliers during semi-supervised learning (SSL) for a standard classifier. In our setting, the central problem is to detect OOD samples from unlabeled wild data and train a dedicated OOD detector. Thus, while SSL methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) seek to improve classification despite outlier contamination, Medix uses detected outliers as positive training evidence for OOD detection in an open-world setting where labeled OOD data is unavailable. Technically, methods such as OpenMatch rely on consistency regularization across stochastic transformations of the same input, implemented through soft constraints on classifier outputs. Medix instead relies on median-based filtering in gradient space, using the robustness of the median to identify candidate OOD samples and deriving error bounds governed by contamination and concentration effects.

B Broader Impacts and Limitations

OOD detection is an important component of reliable model deployment because deployed systems may encounter inputs that differ substantially from training data. Medix contributes a median-centric framework for using unlabeled wild data to improve OOD handling under the study’s stated assumptions. The approach is relevant to safety-sensitive domains such as healthcare, autonomous systems, and finance, where detecting inputs outside the modeled distribution can reduce inappropriate automated decisions. Future work could integrate Medix with generative models for outlier synthesis or adapt the framework to dynamic environments in which OOD distributions evolve over time.

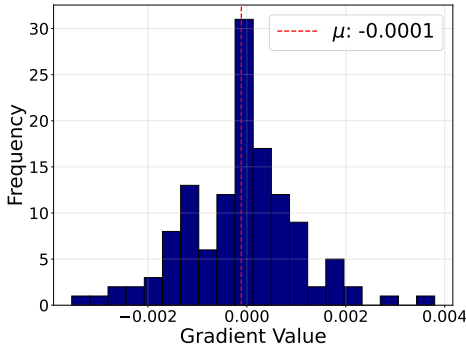
A limitation of Medix is that it assumes access to an unlabeled mixture containing both InD and OOD data; in practice, such data may be noisy, corrupted, or inconsistently sampled. This assumption is not unique to Medix, however, and reflects the uncertainty that motivates robust learning from wild data. We make the relevant conditions explicit in the main theorems, including the geometric separation condition discussed in Remark 4.3, and we also provide a weaker-distributional version in Theorem C.3 that removes the sub-Gaussian tail assumption at the cost of degraded rates. These qualifications delimit the scope of the current guarantees while identifying interpretable directions for extending median-based OOD detection under more realistic noise conditions.

701 C Proof of Theorem

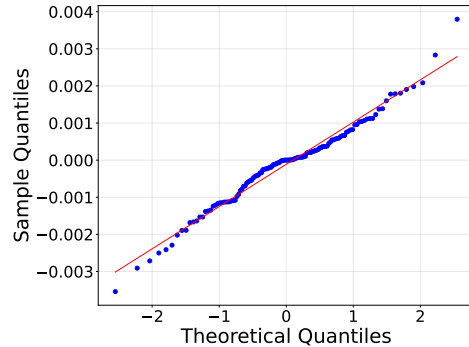
702 For clarity and consistency across the theoretical and empirical sections, Table 7 summarizes the
 703 notation used throughout the paper. The table defines gradient-related quantities, sample sizes,
 704 filtering terms, and the parameters appearing in the concentration bounds. Unless stated otherwise,
 705 boldface denotes vectors, $\mathcal{X}$ denotes the input space, and all gradients are computed with respect to a
 706 fixed pretrained model f_ϕ .

Table 7: Notation summary. We use bold symbols for vectors and $\mathcal{X}$ to denote the input space.

Symbol	Meaning
$\mathcal{X}$	Input space (e.g., images in $[0, 1]^d$)
f_ϕ	Model parameterized by ϕ
$\ell(f_\phi(x))$	Loss function evaluated at input x
$\nabla \ell(f_\phi(x))$	Gradient of loss at point x
$\mathbf{g}(x)$	Shorthand for $\nabla \ell(f_\phi(x))$, gradient vector
$\bar{\mathbf{g}}_{\text{in}}$	Empirical mean of gradients over InD points
m	Total number of unlabeled samples in $\mathcal{S}_{\text{wild}}$
$m_{\text{in}} = (1 - \pi)m$	Number of InD samples
$m_{\text{out}} = \pi m$	Number of OOD samples
$\pi \in (0, 1)$	OOD contamination ratio
$\mathcal{S}_{\text{wild}}$	Unlabeled mixture of InD and OOD data
$\mathcal{S}_{\text{in}}^*$	Inliers selected by the median filter
$\mathcal{S}_{\text{out}}^*$	OOD points selected by the median filter
$\mathcal{B}_\varepsilon(\bar{\mathbf{g}}_{\text{in}})$	Ball of radius ε centered at $\bar{\mathbf{g}}_{\text{in}}$
m_ε	Number of InD points inside $\mathcal{B}_\varepsilon$
m_ε^*	Number of selected inliers inside $\mathcal{B}_\varepsilon$
m_{sep}^*	Number of well-separated OODs in selected set
ERR_{in}	Fraction of InD points misclassified as OOD
ERR_{out}	Fraction of OOD points misclassified as InD
σ^2	Sub-Gaussian proxy variance of each gradient coordinate
ε	Threshold for gradient deviation, $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$
$\delta \in (0, 1)$	Confidence parameter for concentration bounds
d	Dimensionality of gradient vectors



(a) InD gradient histogram.



(b) InD gradient Q-Q plot.

Figure 4: Illustration of InD sample gradients exhibiting sub-Gaussian behavior in each coordinate. (left) Histogram of gradient values (CIFAR-100 InD data) showing concentration around the mean with light tails, consistent with sub-Gaussianity. (right) Q-Q plot comparing empirical quantiles of InD gradients against a theoretical Gaussian distribution, confirming alignment with sub-Gaussianity.

707 **C.1 Bound on the Inlier Misclassification Rate**

708 **Theorem C.1** (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are*
 709 *i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$, and*
 710 *fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification*
 711 *rate of the element-wise median (EWM) filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

712 *Proof.* The proof has two components. In **Step 1**, we derive a high-probability concentration bound
 713 for InD gradients around their mean $\bar{\nabla}_{\text{in}}$ and call an InD sample a *good inlier* when its gradient lies
 714 in this high-probability region. In **Step 2**, we apply a swapping argument to the EWM objective: if
 715 the filter removed more than a $(2\eta + \frac{\pi}{2}(1-\pi))$ fraction of *good inliers*, where η is the fraction of
 716 InD samples outside the high-probability region, then one could construct a competing subset with a
 717 strictly smaller EWM objective value. Combining concentration with this optimality contradiction
 718 gives the stated bound on ERR_{in} .

719 **Step 1: Concentration of InD Gradients.** Let $\nabla \ell(f_{\phi}(\mathbf{x})) \in \mathbb{R}^d$ be the gradient (with respect to ϕ)
 720 of a loss evaluated at an InD sample $\mathbf{x}$. Suppose each coordinate of this gradient is sub-Gaussian with
 721 variance proxy σ^2 , centered at the same mean vector $\bar{\nabla}_{\text{in}}$. That is, for each coordinate $j \in \{1, \dots, d\}$,

$$\mathbb{P}\left(|\nabla \ell(f_{\phi}(\mathbf{x}))_j - \bar{\nabla}_{\text{in},j}| \geq \epsilon\right) \leq 2 \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (6)$$

722 Define

$$\mathbf{v} := \nabla \ell(f_{\phi}(\mathbf{x})) - \bar{\nabla}_{\text{in}} \in \mathbb{R}^d.$$

723 It is straightforward to show that

$$\|\mathbf{v}\|_2 \leq \sqrt{d} \|\mathbf{v}\|_{\infty},$$

724 so if $\|\mathbf{v}\|_2 > \epsilon \sqrt{d}$ then at least one coordinate j must satisfy $|v_j| > \epsilon$.

725 Hence, combining (6) across $j = 1, \dots, d$ via a union bound yields

$$\mathbb{P}\left(\|\nabla \ell(f_{\phi}(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (7)$$

726 Let $m_{\text{in}} = (1 - \pi)m$ denote the total number of InD points in the set $\mathcal{S}_{\text{wild}}$. We define an InD point
 727 $\mathbf{x}$ to be *bad* if

$$\|\nabla \ell(f_{\phi}(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}. \quad (8)$$

728 The objective of this step is to show that, with high probability, only a small fraction of the m_{in} inliers
 729 are bad.

730 The sub-Gaussian assumption and the union bound over the d coordinates define the following upper
 731 bound $p \in [0, 1]$:

$$p := \mathbb{P}\left(\|\nabla \ell(f_{\phi}(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (9)$$

732 Here, $\mathbf{x}$ denotes a generic InD sample, and the m_{in} inliers are assumed to be i.i.d.

733 Let X be the random variable denoting the number of bad inliers among the m_{in} samples. For each
 734 inlier $\mathbf{x}_i$, define the indicator variable

$$I_i := \begin{cases} 1, & \text{if } \|\nabla \ell(f_{\phi}(\mathbf{x}_i)) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

735 Then X is

$$X = \sum_{i=1}^{m_{\text{in}}} I_i. \quad (11)$$

736 Linearity of expectation gives

$$\mathbb{E}[X] = \sum_{i=1}^{m_{\text{in}}} \mathbb{E}[I_i] \quad (12)$$

$$= \sum_{i=1}^{m_{\text{in}}} \mathbb{P}(\mathbf{x}_i \text{ is bad}) \quad (13)$$

$$\stackrel{\text{from (9)}}{\leq} m_{\text{in}} \cdot p \quad (14)$$

$$\leq m_{\text{in}} \cdot 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (15)$$

737 Since X is a sum of m_{in} Bernoulli(p) random variables bounded in $[0, 1]$, Hoeffding's inequality
738 yields, for any $t > 0$,

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (16)$$

739 Setting

$$t := \sqrt{\frac{m_{\text{in}} \log\left(\frac{1}{\delta}\right)}{2}}, \quad (17)$$

740 we obtain

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (18)$$

741 Therefore, with probability at least $1 - \delta$,

$$X \leq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (19)$$

742 Dividing by m_{in} gives the desired upper bound on the fraction of bad inliers:

$$\frac{X}{m_{\text{in}}} \leq \frac{\mathbb{E}[X]}{m_{\text{in}}} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (20)$$

$$\stackrel{\text{from (12)}}{\leq} 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (21)$$

$$= \underbrace{\frac{1}{m_{\text{in}}}}_{\text{Concentration term}} + 2 \underbrace{\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Contamination term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}. \quad (22)$$

743 Denote this fraction by η . With $\epsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$, with probability at least $1 - \delta$, no more
744 than an $O\left(d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{m_{\text{in}}}}\right)$ fraction of inliers have gradients outside the radius $\epsilon\sqrt{d}$
745 around $\bar{\nabla}_{\text{in}}$. These samples are the *bad inliers*; the remaining InD samples are *good inliers*.

746 **Step 2: Contradiction via a Swapping Argument** We next show that if S^* excludes too many
747 InD points, specifically if

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (23)$$

748 then one can construct a new set $\widehat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than the EWM of $\mathcal{S}^*$,
 749 contradicting the optimality of $\mathcal{S}^*$.

750 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 751 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (24)$$

752 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

753 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (25)$$

754 Define the swap size as

$$m'_\varepsilon := \min \{m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (26)$$

755 This quantity is the number of good inliers, namely those inside B_ε , that are excluded from $\mathcal{S}^*$ and
 756 can be swapped into the candidate set.

757 Construct a new set $\widehat{\mathcal{S}}$ by removing m'_ε points from $\mathcal{S}^*$ that lie outside B_ε (OOD points or bad inliers)
 758 and inserting m'_ε good inliers from B_ε in their place. The swap preserves cardinality:

$$|\widehat{\mathcal{S}}| = |\mathcal{S}^*|.$$

759 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (27)$$

760 This quantity measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging gives:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (28)$$

761 Under the contradiction hypothesis,

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (29)$$

762 Subtracting η and multiplying by m_{in} gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (30)$$

763 At most a η fraction of inliers lie outside B_ε , and the OOD fraction among all samples is π . Therefore,
 764 the number of samples outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (31)$$

765 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (32)$$

766 Combining these inequalities yields:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (33)$$

Thus $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (34)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (35)$$

Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (36)$$

so points inside B_ε form a strict majority in $\widehat{\mathcal{S}}$.

Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ denote the j -th coordinate values of all gradients in $\mathcal{S}^*$, and let $\widehat{X}^{(j)} = \{\widehat{x}_1^{(j)}, \dots, \widehat{x}_m^{(j)}\}$ denote the corresponding coordinate values in $\widehat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies outside I_j . Consequently, for some coordinate j^* , the swaps increase the number of values inside I_{j^*} from below $m/2$ to above $m/2$. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\widehat{X}^{(j^*)}) \in I_{j^*}, \quad (37)$$

which implies

$$|\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (38)$$

For every remaining coordinate $j \neq j^*$, the swapped points either leave the majority unchanged or increase it; hence the median distance from μ_j cannot increase and may decrease.

Thus, component-wise:

$$|\text{EWM}_j(G_{\widehat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (39)$$

with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\widehat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (40)$$

This strict improvement contradicts the assumption that $\mathcal{S}^*$ minimizes the optimization objective, completing the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (41)$$

□

C.2 Bound on the Outlier Retention Rate

We next establish a non-asymptotic upper bound on the fraction of OOD points retained by the MEDIX filter. This result complements the inlier misclassification bound (Appendix C.1); instead of relying only on InD concentration, it also uses a separation condition between the OOD and InD gradient means.

Theorem C.2 (Outlier Retention Bound under Vector Separation). *Assume that for every OOD point $x \sim P_{\text{out}}$, the gradient $g(x) := \nabla \ell(f_{\varphi_{\text{in}}}(x)) \in \mathbb{R}^d$ satisfies:*

(I) **Sub-Gaussianity.** *Each coordinate $g_j(x)$ is sub-Gaussian with variance proxy σ_{out}^2 .*

(II) **Vector Mean Separation.** *The mean OOD gradient vector μ_{out} satisfies*

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta \sqrt{d}, \quad \text{for some } \Delta > 0.$$

793 Fix any tolerance $\varepsilon \in (0, \Delta)$ and confidence parameter $\delta \in (0, 1)$. Define:

$$p_{\text{out}}(\varepsilon) := 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right), \quad \eta_{\text{out}} := p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}.$$

794 Then with probability at least $1 - \delta$, the outlier retention rate satisfies:

$$\text{ERR}_{\text{out}} := \frac{|\mathcal{S}_{\text{in}}^* \cap \text{OOD}|}{m_{\text{out}}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}. \quad (42)$$

795 **Proof. Step 1: Concentration of OOD Gradients.**

796 By assumption, each coordinate of $g(x)$ is sub-Gaussian, and the mean vector is separated from $\bar{\nabla}_{\text{in}}$
 797 by at least $\Delta\sqrt{d}$. Therefore, for any OOD point x , the event

$$\|g(x) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \quad (43)$$

798 has probability at most $p_{\text{out}}(\varepsilon)$.

799 **Counting the “ambiguous” OOD points.** For every OOD example x_i define the indicator

$$J_i := \mathbf{1}\left\{\|g(x_i) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}, \quad i = 1, \dots, m_{\text{out}}.$$

800 Each $J_i \sim \text{Bernoulli}(p_{\text{out}}(\varepsilon))$ and the J_i ’s are independent. Let

$$Y := \sum_{i=1}^{m_{\text{out}}} J_i, \quad \mu := \mathbb{E}[Y] = m_{\text{out}} p_{\text{out}}(\varepsilon).$$

801 Because the indicators J_i take values in $\{0, 1\}$, Hoeffding’s inequality states that for any $s > 0$

$$\Pr\{Y - \mu \geq s\} \leq \exp\left(-\frac{2s^2}{m_{\text{out}}}\right).$$

802 Choose $s = \sqrt{\frac{m_{\text{out}} \log(1/\delta)}{2}}$; then (C.2) equals δ , and with probability at least $1 - \delta$

$$Y \leq m_{\text{out}} \left(p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}\right) = m_{\text{out}} \eta_{\text{out}}.$$

803 Define the high-probability event

$$\mathcal{E}_1 := \{Y \leq m_{\text{out}} \eta_{\text{out}}\}, \quad (35)$$

804 which holds with probability at least $1 - \delta$.

805 **Step 2: Contradiction via a Swapping Argument** Suppose, toward contradiction, that

$$\text{ERR}_{\text{out}} > \eta_{\text{out}} + \frac{1 - \pi}{2\pi}. \quad (44)$$

806 Let $m_{\text{out}}^* := |\mathcal{S}_{\text{in}}^* \cap \text{OOD}|$, and define $m_{\varepsilon}^* := \#\{\text{ambiguous OODs in } \mathcal{S}_{\text{in}}^*\}$. Then, under $\mathcal{E}_1$,

$$m_{\text{sep}}^* := m_{\text{out}}^* - m_{\varepsilon}^* > \frac{1 - \pi}{2\pi} m_{\text{out}} = \frac{1}{2} m_{\text{in}}. \quad (45)$$

807 Thus, more than half the points in $\mathcal{S}_{\text{in}}^*$ are separated OODs whose gradients lie outside the ball

$$B_{\varepsilon} := \left\{g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}. \quad (46)$$

By the definition of the coordinate-wise median, this implies that in every coordinate, the majority of entries in $G_{\mathcal{S}_{\text{in}}}^{(j)}$ lie outside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, hence:

$$\|\text{EWM}(G_{\mathcal{S}_{\text{in}}}^*) - \bar{\nabla}_{\text{in}}\|_2^2 \geq \varepsilon^2 d. \quad (47)$$

Now consider a new subset $\hat{\mathcal{S}}$ formed by swapping the m_{sep}^* separated OODs in $\mathcal{S}_{\text{in}}^*$ with an equal number of InD points whose gradients lie inside B_ε . Such inliers exist because fewer than $m_{\text{out}}\eta_{\text{out}}$ are ambiguous, and the total number of InD points is m_{in} .

Then at least half the entries in every coordinate of $G_{\hat{\mathcal{S}}}^{(j)}$ lie inside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, implying:

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2^2 \leq \varepsilon^2 d. \quad (48)$$

Combining with (47) yields a contradiction:

$$\|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\hat{\mathcal{S}}})\|_2^2 < \|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}_{\text{in}}}^*)\|_2^2. \quad (49)$$

This contradicts the optimality of $\mathcal{S}_{\text{in}}^*$, and so the assumption must be false. Therefore, our assumption must be false. With probability at least $1 - \delta$, we conclude that the outlier retention rate is bounded by

$$\text{ERR}_{\text{out}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}, \quad (50)$$

□

C.3 Theorem C.3 without Sub-Gaussianity

Theorem C.3 (Inlier Misclassification Bound without Sub-Gaussianity). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and that each coordinate has variance σ^2 and finite fourth moment bounded by μ_4 , i.e.,*

$$\mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^2 = \sigma^2, \quad \mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^4 \leq \mu_4.$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the number of InD samples in $\mathcal{S}_{\text{wild}}$, and fix any tolerance $\varepsilon > \sigma$ and confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the element-wise median (EWM) filtering rule satisfies:

$$\text{ERR}_{\text{in}} \leq 2 \underbrace{\left(\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \right)}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1 - \pi)}}_{\text{Contamination term}}.$$

Differences with the original bound. The original bound, which assumes sub-Gaussianity, features a decay term

$$2 \exp\left(-\frac{\varepsilon^2 d}{2\sigma_{\text{sub}}^2}\right)$$

that decreases exponentially with dimension d , making the bound extremely tight for large d . In contrast, the new bound, relying solely on the assumption of finite fourth moments, exhibits a decay term

$$\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2}$$

832 that decays as $O(1/d)$, yielding a looser bound for larger d . The original bound requires sub-Gaussian
 833 tails, while the new one only assumes finite fourth moments, allowing for heavier tails. In particular,
 834 both bounds share the finite-sample term

$$\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}},$$

835 that shrinks as m_{in} increases, independent of the tail assumptions.

836 *Proof.* The proof follows the same two-part structure as the sub-Gaussian result: first, it bounds
 837 the fraction of “bad” InD samples, namely those far from ∇_{in} , without invoking sub-Gaussian tails;
 838 second, it applies the same swapping argument to control the inlier misclassification rate. The only
 839 substantive change is that the sub-Gaussian tail bound is replaced by a moment-based bound derived
 840 from the finite fourth-moment assumption.

841 **Step 1: Concentration of InD Gradients** Define the gradient deviation for an InD sample x as:

$$v = \nabla \ell(f_\phi(x)) - \nabla_{\text{in}} \in \mathbb{R}^d, \quad (51)$$

842 where $\nabla_{\text{in}} = \frac{1}{n} \sum_{(x_i, y_i) \in S_{\text{in}}} \nabla \ell(f_\phi(x_i))$ is the mean gradient over the labeled InD training set S_{in} ,
 843 and we assume ∇_{in} is the population mean. An InD sample is classified as “bad” if:

$$\|v\|_2 > \epsilon \sqrt{d}, \quad (52)$$

844 and “good” otherwise. Our goal is to bound the fraction of bad inliers among the m_{in} InD samples in
 845 S_{wild} .

846 Since $\|v\|_2^2 = \sum_{j=1}^d v_j^2$, we analyze the ℓ_2 norm via the sum of squared coordinates. Assume that
 847 the coordinates $v_j = \nabla \ell(f_\phi(x))_j - \nabla_{\text{in},j}$ are independent across $j = 1, \dots, d$, each with:

$$\mathbb{E}[v_j] = 0 \quad (\text{assuming } \nabla_{\text{in},j} = \mathbb{E}[\nabla \ell(f_\phi(x))_j]), \quad (53)$$

$$\mathbb{E}[v_j^2] = \sigma^2, \quad \mathbb{E}[v_j^4] \leq \mu_4. \quad (54)$$

848 Thus:

$$\mathbb{E}[\|v\|_2^2] = \mathbb{E}\left[\sum_{j=1}^d v_j^2\right] = d\sigma^2, \quad (55)$$

$$\text{Var}(\|v\|_2^2) = \text{Var}\left(\sum_{j=1}^d v_j^2\right) = \sum_{j=1}^d \text{Var}(v_j^2). \quad (56)$$

849 Compute the variance of v_j^2 :

$$\text{Var}(v_j^2) = \mathbb{E}[v_j^4] - (\mathbb{E}[v_j^2])^2 \leq \mu_4 - \sigma^4, \quad (57)$$

850 so:

$$\text{Var}(\|v\|_2^2) \leq d(\mu_4 - \sigma^4). \quad (58)$$

851 Using Chebyshev's inequality for $\|v\|_2^2$:

$$\mathbb{P}(|\|v\|_2^2 - d\sigma^2| \geq t) \leq \frac{d(\mu_4 - \sigma^4)}{t^2}. \quad (59)$$

852 We need $\mathbb{P}(\|v\|_2 > \epsilon\sqrt{d})$, which corresponds to $\|v\|_2^2 > \epsilon^2 d$. Set the threshold:

$$\|v\|_2^2 > \epsilon^2 d \implies \|v\|_2^2 - d\sigma^2 > \epsilon^2 d - d\sigma^2 = d(\epsilon^2 - \sigma^2). \quad (60)$$

853 Thus:

$$\mathbb{P}(\|v\|_2^2 > \epsilon^2 d) = \mathbb{P}(\|v\|_2^2 - d\sigma^2 > d(\epsilon^2 - \sigma^2)) \leq \frac{d(\mu_4 - \sigma^4)}{[d(\epsilon^2 - \sigma^2)]^2} = \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}, \quad (61)$$

854 provided $\epsilon > \sigma$ so that $\epsilon^2 - \sigma^2 > 0$. Define:

$$p = \mathbb{P}(\|\nabla \ell(f_\phi(x)) - \nabla_{\text{in}}\|_2 > \epsilon\sqrt{d}) \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (62)$$

855 Let X be the number of bad inliers among the m_{in} InD samples in S_{wild} , where each InD sample is
856 bad with probability at most p . Then:

$$\mathbb{E}[X] \leq m_{\text{in}}p \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (63)$$

857 Since $X = \sum_{i=1}^{m_{\text{in}}} I_i$, where $I_i = 1$ if the i -th InD sample is bad and 0 otherwise, and I_i are i.i.d.
858 Bernoulli(p) variables bounded in $[0, 1]$, apply Hoeffding's inequality:

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (64)$$

859 Set $t = \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}$, so:

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (65)$$

860 With probability at least $1 - \delta$:

$$X \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (66)$$

861 The fraction of bad inliers is:

$$\eta = \frac{X}{m_{\text{in}}} \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}. \quad (67)$$

862 This η represents the upper bound on the fraction of InD samples whose gradients deviate from ∇_{in}
863 by more than $\epsilon\sqrt{d}$, relying only on finite fourth moments and coordinate independence.

864 **Step 2: Contradiction via a Swapping Argument** We next show that if $\mathcal{S}^*$ excludes too many
 865 InD points, specifically if

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (68)$$

866 then one can construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than the EWM of $\mathcal{S}^*$,
 867 contradicting the optimality of $\mathcal{S}^*$.

868 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 869 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (69)$$

870 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

871 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (70)$$

872 Define the swap size as

$$m'_\varepsilon := \min \{m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (71)$$

873 This quantity is the number of good inliers, namely those inside B_ε , that are excluded from $\mathcal{S}^*$ and
 874 can be swapped into the candidate set.

875 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ε points from $\mathcal{S}^*$ that lie outside B_ε (OOD points or bad inliers)
 876 and inserting m'_ε good inliers from B_ε in their place. The swap preserves cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

877 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (72)$$

878 This quantity measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging gives:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (73)$$

879 Under the contradiction hypothesis,

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (74)$$

880 Subtracting η and multiplying by m_{in} gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (75)$$

881 At most a η fraction of inliers lie outside B_ε , and the OOD fraction among all samples is π . Therefore,
 882 the number of samples outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (76)$$

883 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (77)$$

884 Combining these inequalities yields:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (78)$$

885 Thus $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (79)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (80)$$

886 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (81)$$

887 so points inside B_ε form a strict majority in $\widehat{\mathcal{S}}$.

888 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ denote the j -th coordinate values of
 889 all gradients in $\mathcal{S}^*$, and let $\widehat{X}^{(j)} = \{\widehat{x}_1^{(j)}, \dots, \widehat{x}_m^{(j)}\}$ denote the corresponding coordinate values in $\widehat{\mathcal{S}}$.
 890 Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

891 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
 892 outside I_j . Consequently, for some coordinate j^* , the swaps increase the number of values inside I_{j^*}
 893 from below $m/2$ to above $m/2$. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\widehat{X}^{(j^*)}) \in I_{j^*}, \quad (82)$$

894 which implies

$$|\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (83)$$

895 For every remaining coordinate $j \neq j^*$, the swapped points either leave the majority unchanged or
 896 increase it; hence the median distance from μ_j cannot increase and may decrease.

897 Thus, component-wise:

$$|\text{EWM}_j(G_{\widehat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (84)$$

898 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\widehat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (85)$$

899 This strict improvement contradicts the assumption that $\mathcal{S}^*$ minimizes the optimization objective,
 900 completing the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (86)$$

901

□